

LDV_PHEV

Plug-in Hybrid Light Duty Vehicle (combined cycle)



Plug-In Hybrid

Inputs / Outputs

dir	comm	unit	group
Input	GSL	PJ	i
Input	ELCPJ	PJ	i
Output	PLDVHWY	MPKm	o
Output	PLDVCTY	MPKm	o

Efficiency (ceff)

	comm	use2cact	cact2cout	share.up
3	PLDVHWY	1095.122	2.03	0.4
4	PLDVCTY	809.966	2.03	0.8

Key Parameters

cap2act	12.04
olife	16 years
stock	0 1000 Vehicles
cap unit	1000 Vehicles
act unit	million km, city
cost unit	USD/(Vh*km)

Fuel Efficiency

Mode	Fuel	L/100 km	km/kWh	MJ/km	mpg (US)
Highway	GSL + ELCPJ	2.67	3.94	0.91	88.10
City	GSL + ELCPJ	3.61	2.92	1.23	65.16

Vehicle Parameters

Fuels	GSL, ELCPJ	Primary fuel	GSL
Efficiency (highway)	2.67 L/100km	Efficiency (city)	3.61 L/100km
Annual distance	12,043 km/yr	Capacity unit	1000 vehicles
Load/vehicle (hwy)	2.03 passengers	Load/vehicle (city)	2.03 passengers
Max share (highway)	40%	Max share (city)	80%
Vehicle cost	85,108 USD	Annual fixed cost	25 USD/yr
Lifetime	16 yr		

Capacity bounds

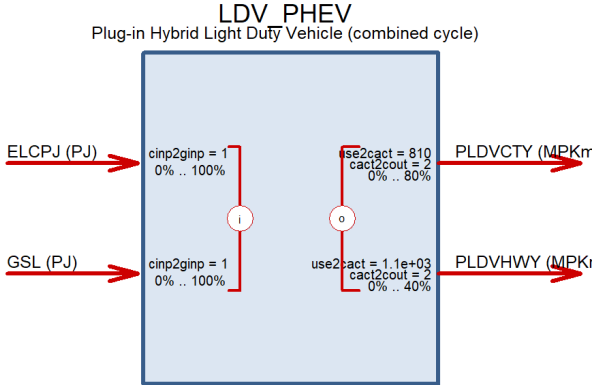
region	year	stock	cap.lo	cap.up	cap.fx	ncap.lo	ncap.up	ncap.fx	ret.lo	ret.up	ret.fx
IND	2024	0	NA	NA	NA	NA	NA	NA	NA	NA	NA

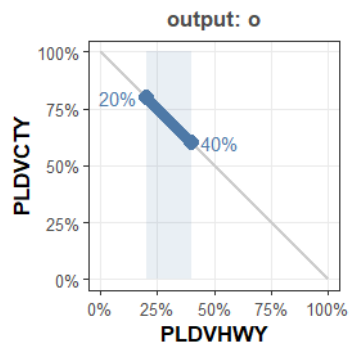
Costs [USD/(Vh*km)]

Fixed O&M	0.02
Investment cost	85.11

Share Mix Constraints

Schematic





Levelized Cost 0.78 [USD/(Vh*km)], 0.39 [USD/(Passenger*km)]

